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Pitch Presentation

TRACK. RIDE. REPEAT.

A campus commute product case study



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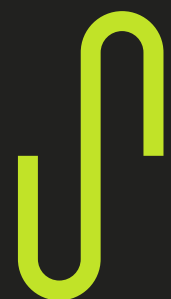
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THE PROBLEM

Students and staff often find the insti bus service unpredictable and confusing.

They don't know where the buses are or when they'll arrive, which leads to long waits and missed rides.

New users also struggle to understand the routes and stops, making it even harder to get around.



USER EMOTIONS



"Two buses towards the same route come one after the other, and then there's nothing for 30–40 minutes? I don't even know if I just missed a bus or if it just hasn't come yet."

"I had heavy luggage, but the bus was packed. I couldn't walk to main gate, so I had to pay extra for a cab to come inside."

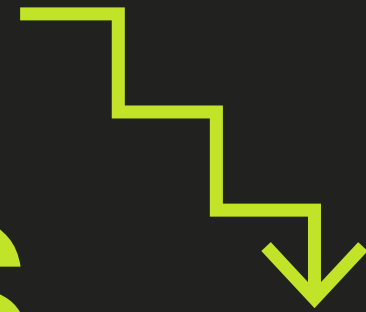


"I was new to insti and didn't know the bus routes properly. I went from Velachery Gate hoping to reach my hostel, but the bus I boarded went to Main Gate instead. I was totally lost."

"I was already nervous for my endsem, and waiting cluelessly at the stop made it worse. There's zero predictability tbh"



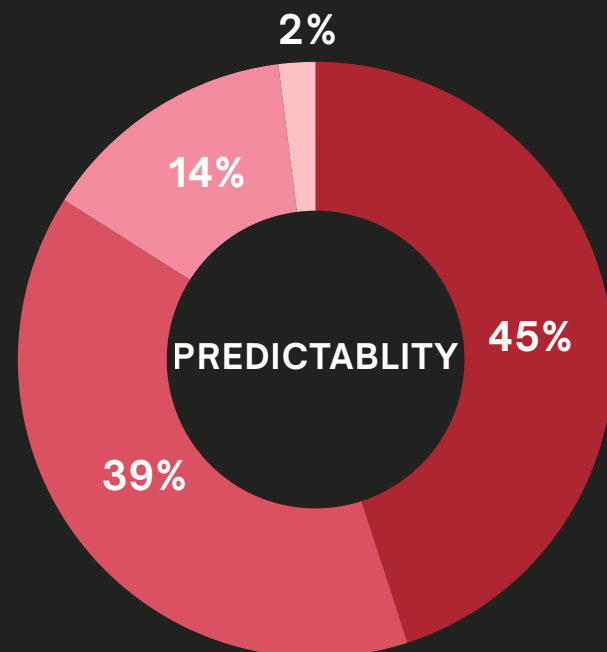
USER INSIGHTS



A survey was conducted amongst insti students and garnered over 60+ responses. It can also be potentially expanded to other users like school students, staff, non-teaching staff, cleaning staffs etc. in the future to get more insights on the same.

► LOW PREDICTABILITY

Buses sometimes come back-to-back, then disappear for a long time.



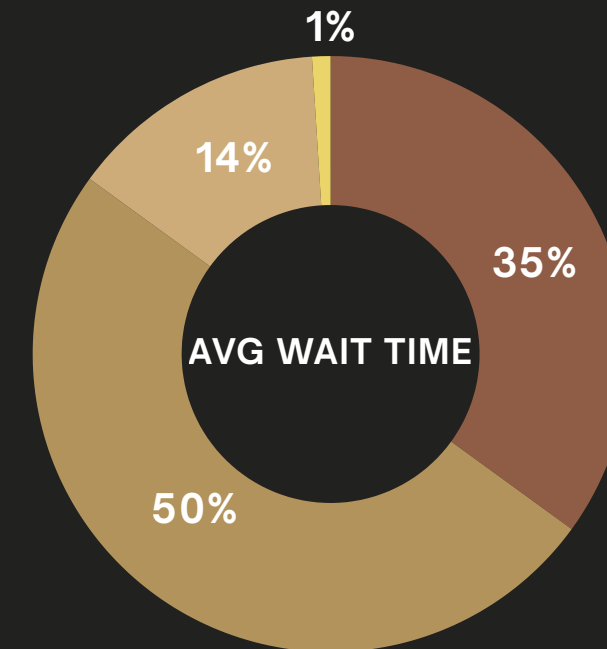
SOMEWHAT PREDICTABLE

MOSTLY UNPREDICTABLE

COMPLETELY RANDOM

VERY PREDICTABLE

► LONG WAITS ARE THE NORM



A majority (85%) of users wait for 5 to 30 minutes, with 50% waiting over 15 minutes.

5 TO 15 MIN

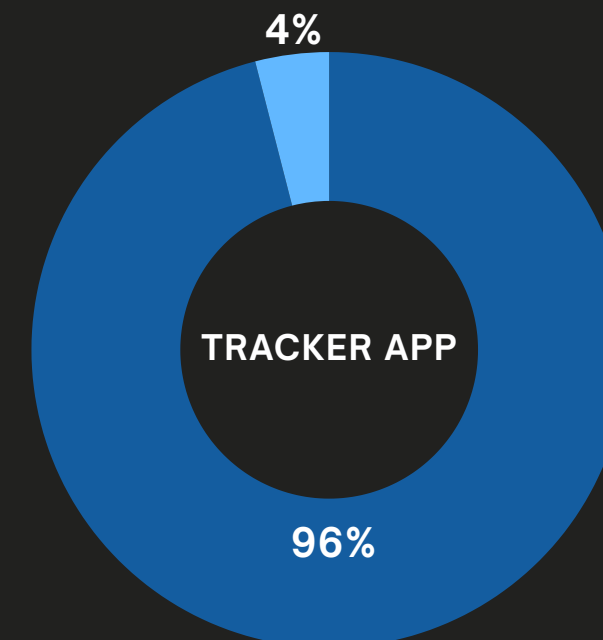
15 TO 30 MIN

MORE THAN 30 MIN

LESS THAN 5 MIN

► STRONG DEMAND FOR A SOLUTION

96% would definitely use a live bus tracker - CLEAR VALIDATION FOR THE PRODUCT IDEA!



DEFINITELY

MAYBE

KEY USER PAIN POINTS



UNPREDICTABILITY & LACK OF VISIBILITY

- Long and Unpredictable wait times
- No real-time bus visibility
- No info if the bus just missed

ACADEMIC AND CRITICAL TIME IMPACTS

- Missed classes or exams due to delays
- Stress during endsems and travel days



ACCESSIBILITY ISSUES AT ODD HOURS

- Overcrowded buses during peak hours
- Poor frequency during afternoons/lunch
- No alternative when buses are full or late
- Auto/cab drivers refuse to enter or charge extra

VOICES FROM THE BUS QUEUE



USER EXPECTATIONS

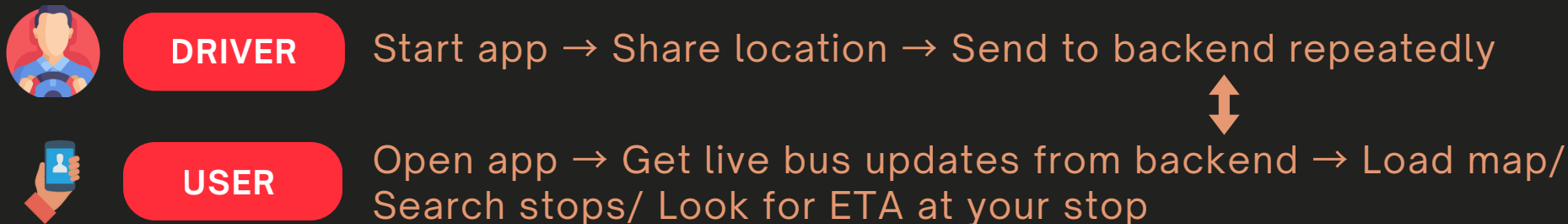
- “Open the app, spot the bus, that’s it”
- “Give me a map and ETA — no big texts, no confusion”
- “I just want to know where the bus is right now”



THE SOLUTION

Live Bus Location Tracker (MVP)

A mobile-first, lightweight app built using Flutter and Firebase, designed to offer real-time bus tracking with minimal setup — powered by either a smartphone or GPS device on the bus.



- ✓ REMOVES UNCERTAINTY
- ✓ REDUCES WAIT-TIME
- ✓ HELPS PLAN DEPARTURES BETTER

FEATURE PRIORITIZATION

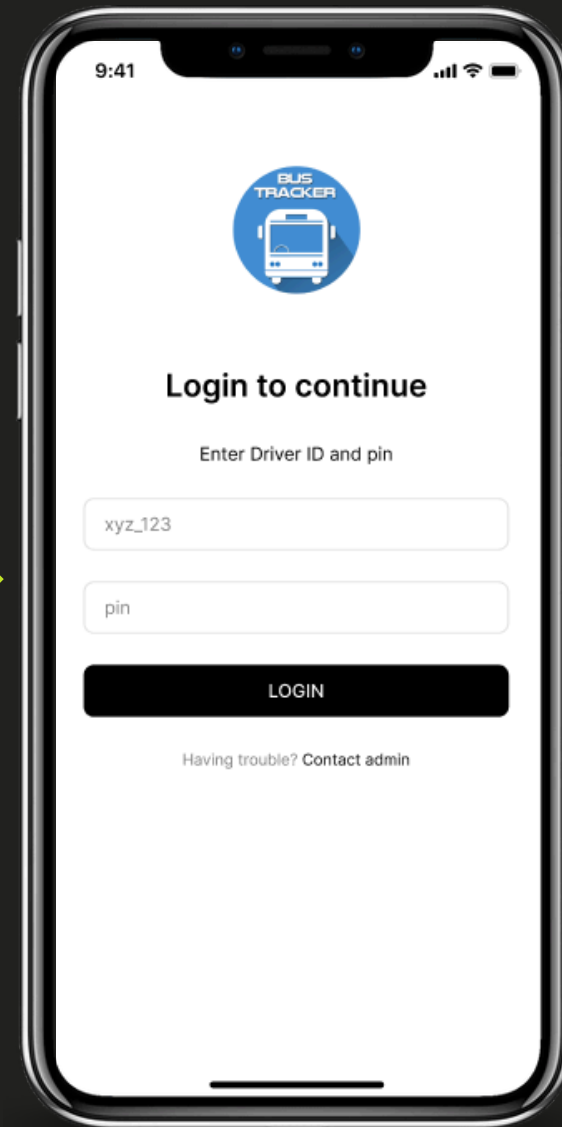
- 1 Live insti map showing all active buses; updated every 5-10 sec as they move
- 2 Movement indicators labelled by directions on live bus icons — eg: “To hostel”
- 3 Nearby Stops and Routes, ETA of buses arriving to that stop
- 4 Search bar (on home page and map page) for searching routes and stops
- 5 No special login required for users, just open the app and start tracking

APP FLOW - DRIVER



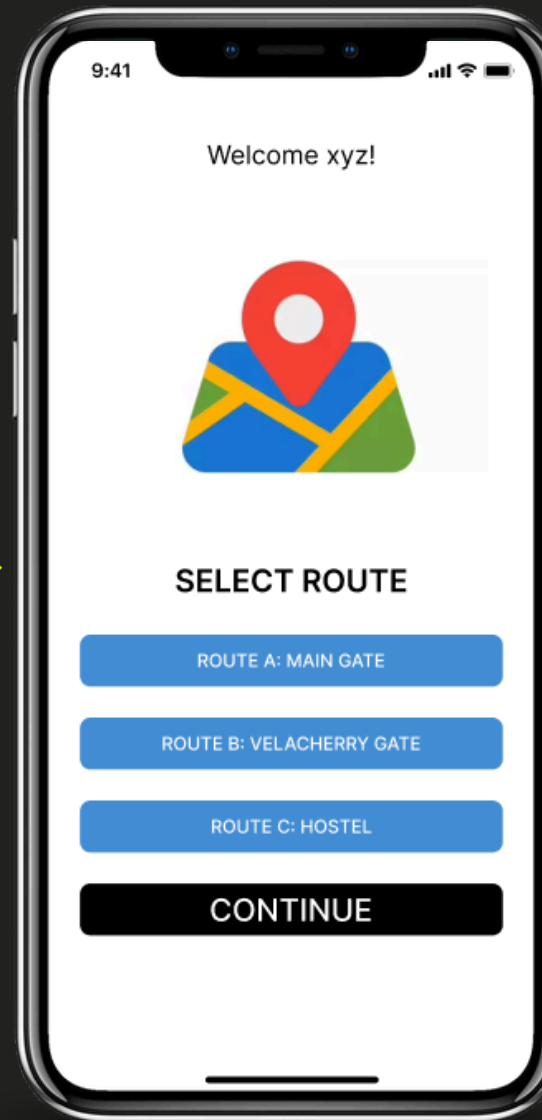
1

SPLASH SCREEN-
ROLE SELECTION



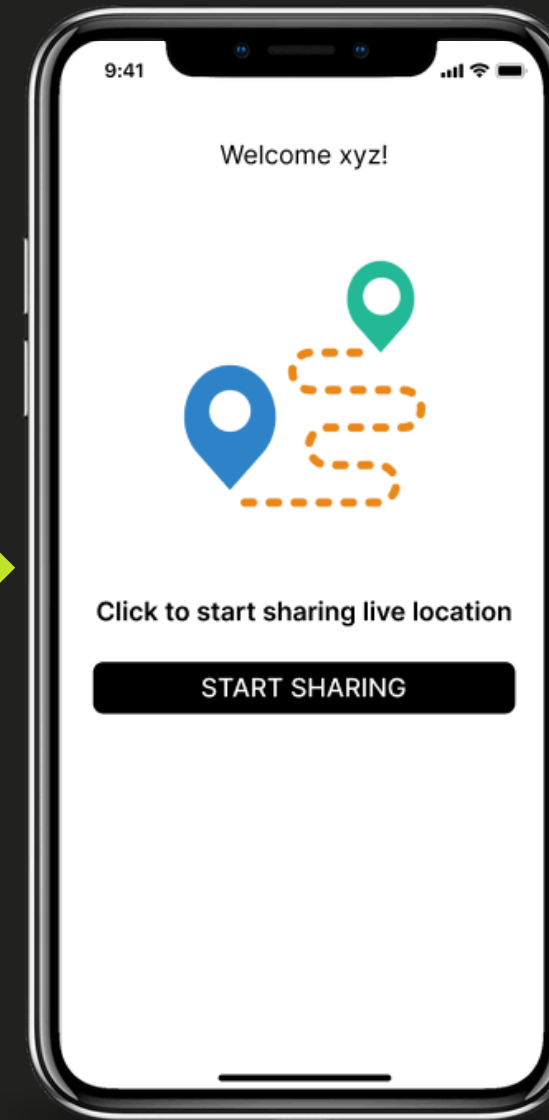
2

DRIVER LOGIN
PAGE



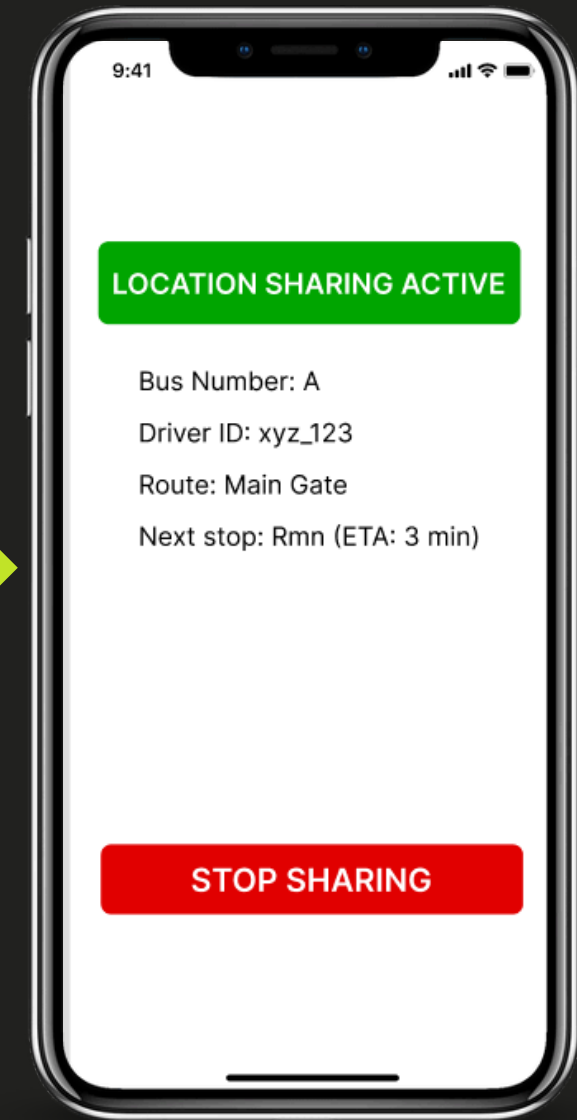
3

ROUTE SELECTION



4

LOCATION
SHARING



5

DISPLAY STATUS &
STOP SHARING

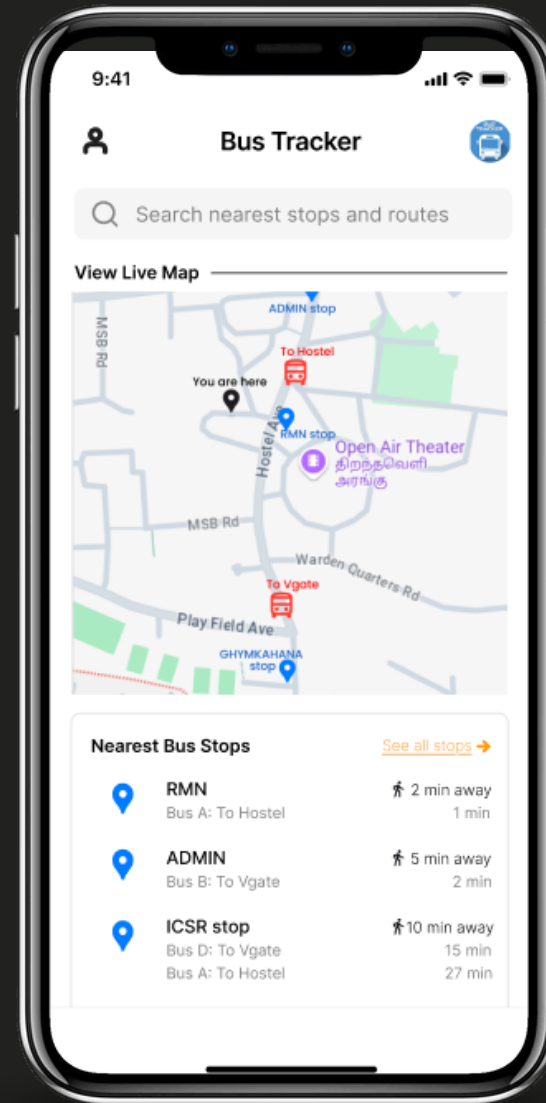
Note: [Figma design Mockup link](#)
[Figma clickable prototype link](#)

APP FLOW - USER



1

SPLASH SCREEN-
ROLE SELECTION



2

USER HOME PAGE
*See nearby stops and
buses, search bar*

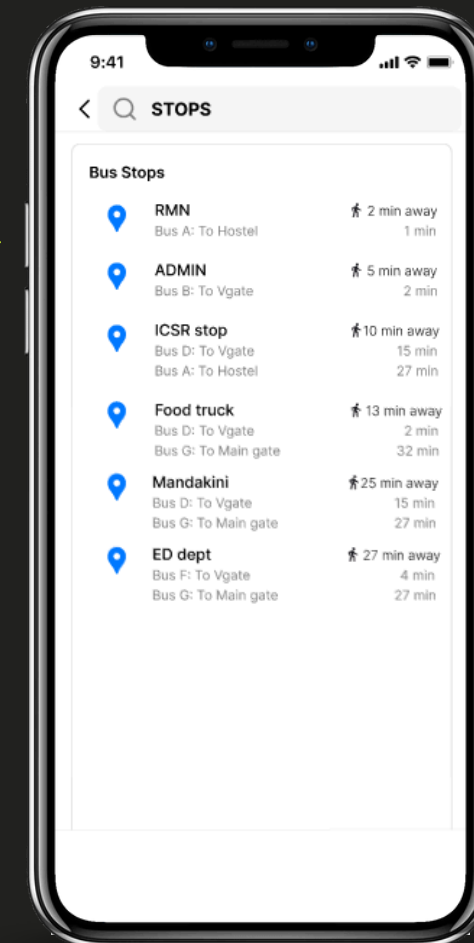
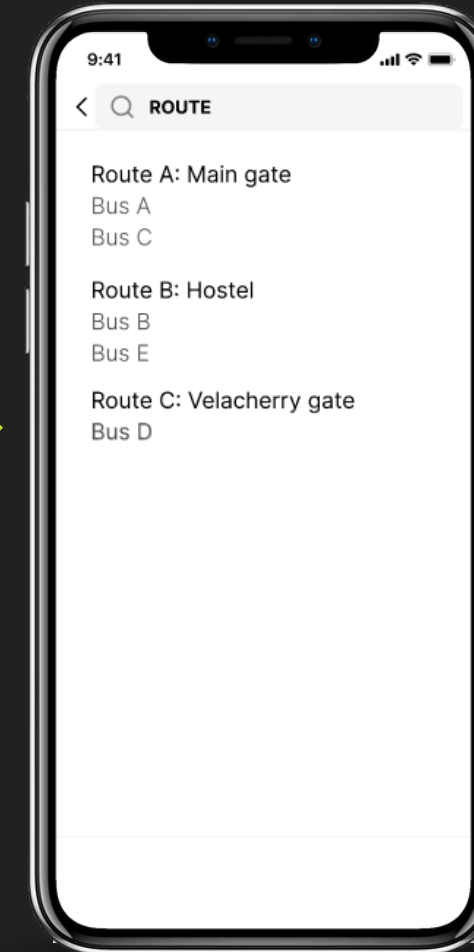


💡 CORE MVP FEATURE:
LIVE INSTI MAP



3

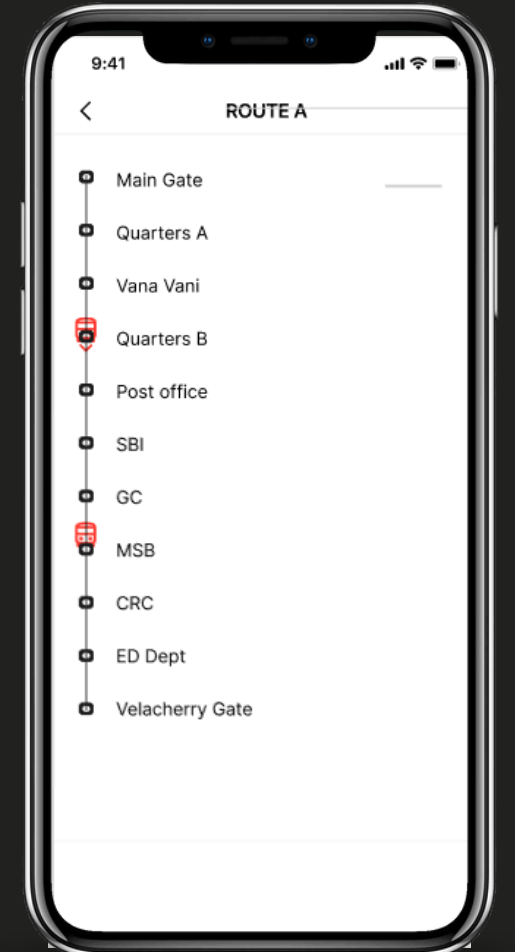
LIVE MAP PAGE
*Live map with active
buses and their route
pop ups, stop info*



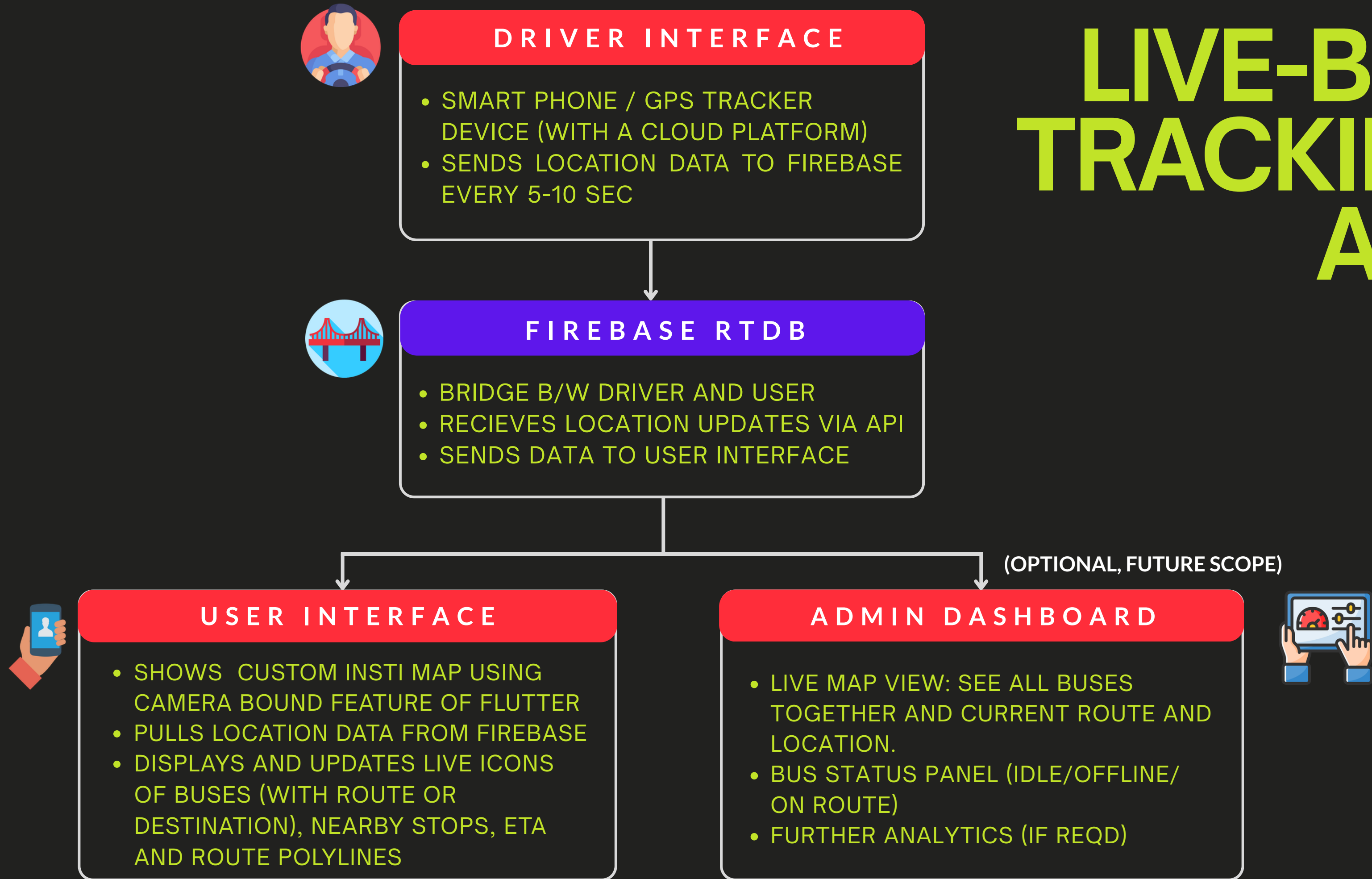
4

SEARCH RESULT
PAGES

*Search for routes,
stops and see active
buses in particular
routes*



LIVE-BUS TRACKING APP



ROADMAP



PHASE 1: RESEARCH & SURVEY

- Identified pain points
- Collected feedback from 50+ users
- Validated the need for a live bus tracking tool

PHASE 2: MVP BUILD

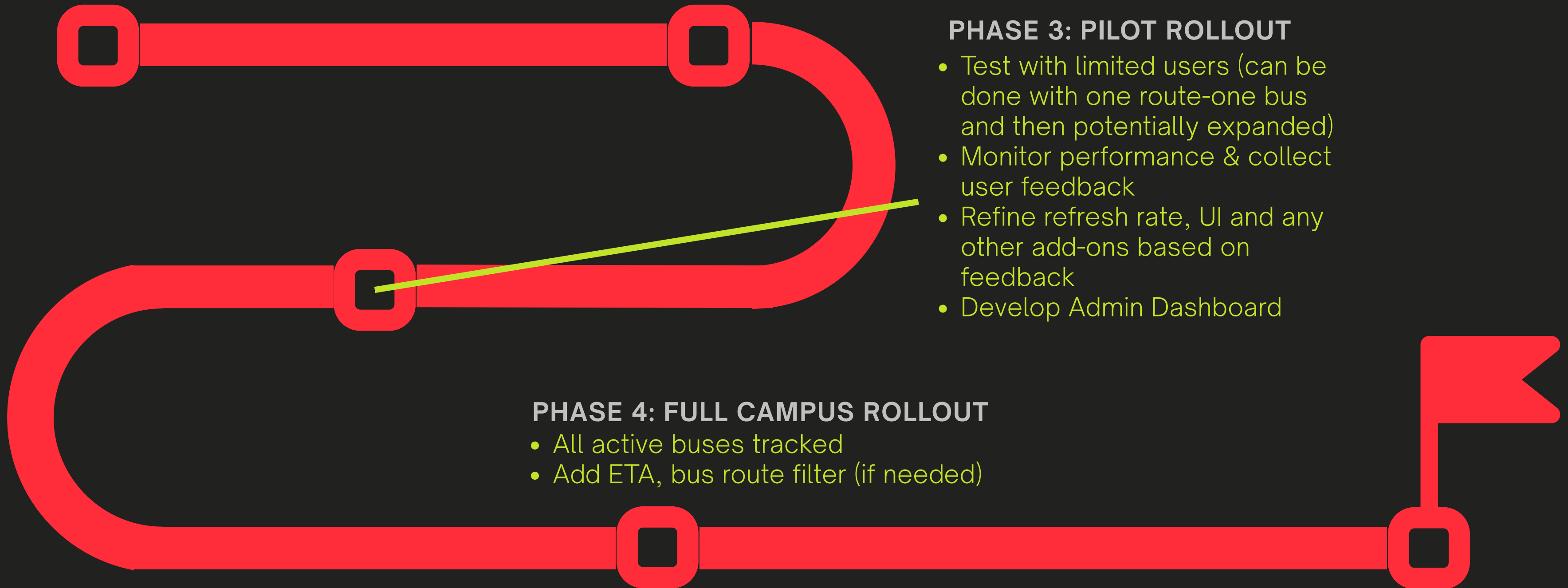
- Show map + real-time bus icons (basic live tracking) + stops, ETA for each bus and stop & route polylines
- 5–6 buses tracked and visible on the map

PHASE 3: PILOT ROLLOUT

- Test with limited users (can be done with one route-one bus and then potentially expanded)
- Monitor performance & collect user feedback
- Refine refresh rate, UI and any other add-ons based on feedback
- Develop Admin Dashboard

PHASE 4: FULL CAMPUS ROLLOUT

- All active buses tracked
- Add ETA, bus route filter (if needed)



IMPACT



1. REDUCED WAIT TIME

Aims to **reduce average wait time by 75%**, bringing it down to **< 5 mins** (current average waiting time is ~19.7 mins based on survey data).



2. IMPROVED TIME MANAGEMENT

Helps students plan commutes more efficiently, freeing up ~15–20 minutes daily for academic or personal use through **predictable ETAs**.



3. INCREASED RELIABILITY & TRUST

Increases reliability on insti bus services by replacing vague or word-of-mouth updates with **real-time location visibility**.

FUTURE SCOPE



Admin dashboard

For managing routes, monitoring live bus status, and resolving issues in real-time.



Delay alerts for users

Notifies users instantly about delays or reschedules.



Feedback collection system

Share feedback to improve service quality of the ride as well as the app functionality.



Offline mode support

Enables basic access to bus info even with poor or no internet (important esp at IITM!)

THANK YOU

